Central Reading of X-rays for Individual Radiographic Features of Hip OA

1. Overview

1.1 SAS datasets

Names: **hxr_sqxx** (xx identifies the time point), hXR_SQ_summary (contains information about prevalence and incidence of radiographic hip OA)
Display label: **hxr SQ reading**

1.2 Contents of dataset

These datasets contains central longitudinal readings of serial pelvis X-rays for radiographic hip OA, and given the OAI project number 21. Readings were performed by by Nancy Lane, MD; Thomas Link, MD; Pia Jungmann, MD and Lorenzo Nardo, MD; under the direction of Dr. Michael Nevitt, PhD of at UCSF.

Each dataset hXR_SQxx contains the reading data for a single visit, or time point. The corresponding longitudinal data for other time points are in separate datasets. For example, the data from the reading of baseline x-rays are in a dataset ending in "00" (i.e., hXR_SQ00), while the corresponding data for the 48-month visit x-ray readings are in a dataset with the same name but ending in "06" (i.e., hXR_SQ06), and so on. Also, most participants had pelvis x-rays acquired at enrollment/baseline visit (V00) and 48-month visit (V06), but a few participants with no V00 x-ray had one acquired at the 12-month visit (V01). The dataset hXR_SQ_summary contains information about prevalence and incidence of radiographic hip osteoarthritis (RHOA) for each participant and side (right and left). (See the "*VisitPrefixDefinitions.pdf*" document for a guide to visit numbering, in the "General Documentation" .zip file, downloadable from http://oai.nih.gov using "Download Documentation" under the "Query and Download" tab) To compare values of a variable across serial time points from a given project by this vendor, or to calculate change scores, users will need to merge the datasets for the various time points. Note that only data from serial time points (i.e. readings of the same side (right or left) from radiographs acquired at different time points) from the same reading project can be used to calculate change between time points.

IMPORTANT: <u>All users are strongly encouraged to read the *Overview and Description of Central Image Assessments* document carefully before attempting to use this dataset!</u>

For comprehensive and detailed information about the structure and contents of the central image assessment datasets, general guidelines for defining change between time points and duration of follow up, and strategies for merging datasets for analysis please see the ImageAssessmentDataOverview.pdf file in "Image Assessment Documentation" .zip file downloadable from http://oai.nih.gov using "Download Documentation" under the "Query and Download" tab. This document is also located in the compressed documentation file (.zip/.exe) that accompanies the corresponding compressed dataset file (.zip/.exe).

1.3 Condition

- Known data errors: problems/cautions for use are listed by variable in the "Release Comments", which can be found in the various h*XR_SQxx_Comments.pdf* files which are located in the compressed documentation (.zip/.exe) that accompanies the corresponding dataset file. If no file exists, there is nothing to report.
- Dataset strengths/weaknesses:
  - Data are expected for all participants included in a project sample. If expected data do not exist for a hip, SAS special missing values are assigned to denote why the data were not obtained.
  - The dataset only contains a single row of data (1 record) for a given hip side (right or left), but other OAI datasets may contain multiple records per side and this needs to be taken into account when merging with other datasets. Please see the ImageAssessmentDataOverview.pdf file in "Image Assessment Documentation" .zip file downloadable from http://oai.nih.gov using "Download Documentation" under the "Query and Download" tab for more information on merging.
  - To examine longitudinal changes in SQ readings for a specific project, records for that project in the hXR_SQ00 (or hXR_SQ01 if no pelvis x-ray at V00) datasets are used for baseline visit values, and then longitudinal changes from those values are found by using values from the same variable in the follow-up hXR_SQ06 dataset which have the same values for ID, SIDE and READPRJ (although for the current hXR_SQxx datasets, READPRJ only takes a single value).
  - The hXR_SQxx dataset should be be merged with the hXR_SQ_summary dataset by ID, SIDE, READPRJ to obtain information about hip OA reading status
  - Definitions of radiographic hip OA used can be found in section 2.3.5.

1.4 Variables and reading methods

Table 1 in Section 2.3 of this document gives a summary of the data and variables in these datasets, but the dataset documentation files *hXR_SQxx_Contents.pdf* and *hXR_SQ_Summary_Contents.pdf* in the compressed documentation file contain a complete list of all the variables in the dataset, their SAS variable names, descriptive variable labels and attributes.

Variables from Project 21 include:

- Individual radiographic features such as osteophytes and joint space narrowing in specific anatomic locations, based on a published atlas(1).
- Radiographic hip OA (RHOA) status calculated from the individual features (see section 2.3.5).
- Meta-data related to reading procedures, exclusions (see section 2.4).

2. Methods specific to Project 21

2.1 Image type

Anterior-posterior pelvis radiographs. The acquisition protocol can be downloaded as a PDF file at the "Operations Manual" link from http://oai.nih.gov using "Download Documentation" under the "Query and Download" tab.

2.2 Time points

Baseline or 12-month, and 48-month visits.

- Most participants have their first pelvis x-ray at the Enrollment visit (V00), but a small subset had their first pelvis x-ray acquired at their 12-month follow-up visit (V01).
- If follow-up pelvis x-rays were acquired and read, they were from the 48-month follow-up visit (V06).
- Many participants in OAI also have a pelvis x-ray acquired at their 96-month follow-up visit (V10) but they were not read as part of this reading project.

2.3 Measurement methods

Radiographs were assessed for individual radiographic features of hip OA using the OARSI atlas(1) and grades by two musculoskeletal radiologists (TML, PMJ, NL) and a rheumatologist (NEL). A list of the assessments are in Table 1. The readings consisted of a screening phase and, for subjects selected during the screening, a complete feature assessment on the available radiographs.

Table 1. Individual Radiographic Features

Individual Radiographic Feature	Variable name in dataset	Grading
Supero-lateral joint space narrowing (JSN)	VxxHJSNSL	0, 1, 2, or 3
Supero-medial joint space narrowing (JSN)	VxxHJSNSM	0, 1, 2, or 3
Superior acetabular osteophytes	VxxHAOSS	0, 0.1+, 1, 2, or 3
Inferior acetabular osteophytes	VxxHAOSI	0, 1, 2, or 3
Superior femoral osteophytes	VxxHFOSS	0, 0.5*, 1, 2, or 3
Inferior femoral osteophytes	VxxHFOSI	0, 0.5*, 1, 2, or 3
Acetabular subchondral cysts	VxxHCYA	absence or presence
Femoral subchondral cysts	VxxHCYF	absence or presence
Acetabular subchondral sclerosis	VxxHSCA	absence or presence
Femoral subchondral sclerosis	VxxHSCF	absence or presence
Femoral head flattening/deformity	VxxHFLAT	absence or presence
+ The non-integer grade of 0.1 represents a reading from the screening process and represents the fact that a grade 0 or a grade 1 superior acetabular osteophyte was present. *The non-integer grade of 0.5 is not described in the referenced atlas, but it was allowed to be scored for femoral osteophytes in order to capture possible (but not definite) osteophytes.		

All readings were performed with the baseline and 48-month radiographs viewed together and in known chronological order.

Participants with acceptable quality radiographs at the baseline and/or 48 month time-point were included in the reading. When only a baseline or only a 48 month radiograph was available, the single time-point was read. A small number of subjects and/or hips with available radiographs are excluded due to the presence of conditions or artifacts that interfere with the reading and/or the interpretation of the findings.

2.3.1 Screening reading

Because of the very large number of subjects in the study and expected presence of RHOA in a relatively small percent, the reading include an initial screening step to identify subjects with possible OA for an intensive assessment of hip OA features. The baseline, and when available, 48-month pelvis radiographs of 4761 participants were screened by one reader for the presence of hip osteophytes (femoral osteophyte OARSI grade >=1 or acetabular osteophyte OARSI grade >= 2) or JSN (OARSI grade >=1) in any location. If any of these features were present, or other OA related features such as femoral head flattening, subchondral sclerosis cysts were recorded, at any time-point the subject was classified as having screened positive for hip OA findings. A second reader then screened the radiographs of subjects without either of these findings at any time-point. If confirmed "negative" by the second reader, the subject was categorized as a "negative screen" result and their radiographs were not evaluated further. The radiographs of 1441 (30%) participants were screened positive for hip OA findings.

2.3.2 Full reading

The radiographs of participants who screened positive for hip OA findings were independently scored for all features listed in Table 1 (both hips, all available time-points) independently by two readers using the OARSI reference atlas[1]. Disagreements between readers for the presence of definite supero-lateral or supero-medial JSN, definite femoral or acetabular osteophytes, presence/absence of any of cysts, sclerosis or femoral head deformity were adjudicated by 3 readers (TML, PMJ, NL) and required the agreement on the score be at least 2 of the 3 readers.

2.3.3 Non-integer grades for osteophytes and JSN

Non-integer grades are not described in the referenced atlas for feature scoring, but were allowed in some circumstances. Superior and inferior femoral osteophytes could be scored as 0.5 to record the presence of very small or questionable possible osteophytes. Within-grade changes in supero-lateral or supero-medial JSN were also scored at follow up visit when the JSN had worsened since the prior visit, but not enough to warrant a whole OARSI grade change.

2.3.4 Score Imputations for normal hips

As outlined about, radiographs which were screened normal by 2 readers have scores of 0 for all features listed in Table 1, **except** for superior acetabular osteophytes which are only known to be grade 0 or 1. For such hips a special value of 0.1 is used in the datasets for the score for superior acetabular osteophytes to indicate this situation.

2.3.5 Definitions of RHOA

Femoral and acetabular osteophytes and supero-lateral and supero-medial JSN with grade >=2 were considered "definite" findings. Hips were classified as "definite RHOA" when:

(1) the modified Croft grade[2] was 2 or greater (presence of two or more of definite osteophytes, definite JSN, sclerosis, cysts or deformity); or
(2) there was definite JSN plus grade >=1 femoral or grade >=2 acetabular osteophytes; or
(3) there was grade >=2 femoral osteophytes regardless of other features; or
(4) there was supero-lateral JSN >=2 or supero-medial JSN>=3 regardless of other features.

Hips were classified as "possible RHOA" when other individual or combinations of, indefinite features were present (e.g., grade 1 osteophytes or grade 1 JSN). All other hips were classified as radiographically "normal"[2,3].

In the hxr_sq_summary dataset, the follow-up variables define RHOA status

V00HIP_STATUS – baseline RHOA status (0: No OA, 1: Possible OA, 2: Definite OA)

V06HIP_STATUS – 48-month RHOA status (0: No OA, 1: Possible OA, 2: Definite OA)

If images were of poor quality, a special missing value .T was used to indicate that situation. The following SAS code will apply the correct format to these variables:

```
proc format library=work; value hipoaiff
0="0: Hip No OA"
1="1: Hip Possible OA"
2="2: Hip Definite OA"
.J=".J:excluded for medical reason"
.P=".P:HR"
.T=".T:poor quality"
.M=".M:missing"
.U=".U:undetermined"
;
run;
```

Poor quality judged when at least two readers (at either screening or adjudicated reading) determined that image quality for a hip made it impossible to determine the grades of the various radiographic features.

2.3.6 Missing Data

Missing data can occur for a variety of reasons. In the individual datasets, SAS special missing values are used to indicate the following:

- .P if data is missing due to a prosthesis/hip replacement.
- .T if data is missing due to technical reasons (e.g. poor image quality).
- .A if the data is not expected (e.g. no x-ray acquired)
- .J for exclusion due to medical conditions (e.g.: Paget's disease, hip fracture, rheumatoid arthritis)

2.4 Variables

Variables are prefixed VxxHXR, and RHOA and OARSI grades are available for all participants in this project at all available time points. Features evaluated were supero-medial and supero-lateral joint space narrowing (JSN), medial and lateral superior and inferior femoral and acetabular osteophytes, subchondral sclerosis, subchondral cysts and femoral head deformity. In the hXR_SQ_summary dataset, there are also meta-data about whether (a) the results from the screening process, or (b) the full reading process, as well as variable giving the RHOA status at baseline and follow-up (for people with radiographs at the 48-month visit).

2.5 Sample

Project 21 contains data for all OAI participants for whom pelvis x-rays were acquired at baseline, 12-month or 48-month visits. The following table gives some demographic information about the participants with data currently released for Project 21:

Table 2a. Project 21 sample: Distribution of Race by Sex, for participants with defined RHOA status

	White or Caucasian	Non-White	Total
Male	1356	287	1823
Female	1937	576	2513
Total	3473	863	4336
5 participants did not specify their race			

Table 2b. Project 21 sample: Distribution of Age by Sex, for participants with defined RHOA status

	Age (years)				
	45 to 49	50 to 59	60 to 69	70 to 79	Total
Male	244	688	456	4381826	1826
Female	269	821	873	552	2515
Total	513	1509	1329	990	4341

2.6 Evaluation of the reading procedures
Radiographs of 140 randomly selected hips that were screened negative for RHOA were blindly read using the study protocol.

The specificity of screening for the absence of definite RHOA was 99.2%, and the estimated sensitivity for definite RHOA was 93%.

Test-retest reliability of the readings was evaluated in a random sample of 189 participants, the radiographs of whom were reread blindly by two readers with adjudication of disagreements. Reliability was generally good for medial and lateral joint space narrowing (weighted κ: 0.72 to 0.77); superior and inferior femoral osteophytes (0.70 to 0.81); acetabular osteophytes (0.51 to 0.64); and cysts, sclerosis, or deformity (0.53 to 0.55) – (see tables 3a, b and c for detailed cross-tabulations)

Reliability for the three level summary classification of RHOA (none, possible, definite) was good (weighted κ: 0.72), as was classification of presence or absence of definite radiographic hip OA (κ: 0.77) – see table 3a.

Table 3a. Agreement between 1st and 2nd reading for Radiographic hip OA and Joint Space Narrowing

RHOA (0=No, 1=Possible, 2=Definite)

		2nd read		
		0	1	2
1st read	0	107	31	4
	1	16	86	13
	2	1	23	95

Supero-medial JSN (OARSI Grade)

		2nd read			
		0	1	2	3
1st read	0	203	32	1	0
	1	14	47	19	0
	2	0	5	48	3
	3	0	0	0	4

Supero-lateral JSN (OARSI Grade)

		2nd read			
		0	1	2	3
1st read	0	252	31	3	0
	1	11	0	11	0
	2	0	7	26	0
	3	0	0	0	5

Table 3b Agreement between 1st and 2nd reading for osteophytes

Superior femoral ostephyte (OARSI Grade)

		2nd read				
		0	0.5	1	2	3
1st read	0	179	38	2	0	0
	0.5	4	20	7	0	0
	1	0	13	52	2	0
	2	0	0	16	37	2
	3	0	0	0	1	3

Superior acetabular ostephyte (OARSI Grade)

		2nd read			
		0	1	2	3
1st read	0	80	44	2	0
	1	21	129	16	0
	2	0	17	49	5
	3	0	0	8	5

Inferior femoral ostephyte (OARSI Grade)

		2nd read				
		0	0.5	1	2	3
1st read	0	298	10	12	0	0
	0.5	1	10	8	0	0
	1	1	6	17	3	0
	2	0	0	4	5	0
	3	0	0	0	0	0

Inferior acetabular ostephyte (OARSI Grade)

		2nd read			
		0	1	2	3
1st read	0	285	31	2	0
	1	12	34	1	0
	2	2	5	4	0
	3	0	0	0	0

Table 4. Agreement between 1st and 2nd reading for subchondral cysts/sclerosis

Femoral subchondral cysts

		2nd read	
		No	Yes
1st read	No	351	20
	Yes	1	6

Acetabular subchondral cysts

		2nd read	
		0	1
1st read	No	344	16
	Yes	3	14

Femoral subchondral sclerosis

		2nd read	
		No	Yes
1st read	No	321	30
	Yes	2	24

Acetabular subchondral sclerosis

		2nd read	
		No	Yes
1st read	No	295	33
	Yes	13	37

3. References

1. Altman RD, Gold GE. Atlas of individual radiographic features in osteoarthritis, revised. Osteoarthritis Cartilage 2007;15 Suppl A:A1-56. PMID: 17320422 http://dx.doi.org/10.1016/j.joca.2006.11.009
2. Croft P, Cooper C, Coggon D. Case definition of hip osteoarthritis in epidemiologic studies. J Rheumatol, 1994, 21(4), 591-592. PMID: 8035380.
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